

PRAKTIKUM SISTEM OPERASI

MODUL 10

SHELL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.

Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

- a. Pada prototypes.h chname diubah menjadi:

```
extern status ascdade(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);

/* in file clkupdate.S */
extern uint32 clkcount(void);
```

- b. Pada chname.c fungsi diubah dari:

```
/*-----
 * chname - Dummy syscall
 *-----
 */
syscall chname(
    pid32      pid,
    char *     newname
)
```

menjadi

```
#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
pri16 chname(
    pid32      pid,
    char *     *newname,
    uint32     namelen
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil

```

/*
 * .....
 */
pril6 chname(
    pid32      pid,
    char       *newname,
    uint32     namelen
)
{
    intmask mask;
    struct procent *prptr;
    uint32 i;

    mask = disable();

    /* Validasi PID */
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];

    /* Nama baru */
    for (i = 0; i < namelen && i < PNMLEN-1; i++) {
        prptr->prname[i] = newname[i];
    }

    /* Null terminator */
    prptr->prname[i] = '\0';

    restore(mask);
    return OK;
}

```

2. Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

- a. Edit shell.c

```

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/*
 * .....
 */
/* Table of Xinu shell commands and the function associated with each */
/*
 * .....
 */
const struct cmdent cmdtab[] = {
    {"argecho", xsh_argecho},
    {"arp", xsh_arp},
    {"cat", xsh_cat},
    {"clear", xsh_clear},
    {"date", xsh_date},
    {"devdump", xsh_devdump},
    {"echo", xsh_echo},
    {"help", xsh_help},
    {"ls", xsh_ls},
    {"kill", xsh_kill},
    {"memdump", xsh_memdump},
    {"memstat", xsh_memstat},
    {"ns", xsh_ns},
    {"netinfo", xsh_netinfo},
    {"ping", xsh_ping},
    {"ps", xsh_ps},
    {"sleep", xsh_sleep},
    {"tee", xsh_tee},
    {"udp", xsh_udpdump},
    {"udpecho", xsh_udpecho},
    {"udpserver", xsh_udpserver},
    {"uptime", xsh_uptime},
    {"namecmd", xsh_namecmd},
    {"mycmd", xsh_mycmd},
    {"?", xsh_help}
}

```

- b. Edit shprototypes.h

```

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);

```

- c. Buat file baru:

```
#include <xinu.h>
#include <stdio.h>
#include <string.h>

shellcmd xsh_namecmd(int32 nargs, char *args[])
{
    printf("My new command\n");

    /* pid 3 = neti, ubah nama proses menjadi hello */
    chname(2, "hello", 5);

    return 0;
}
```

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3. Test hasilnya:
- a. Masuk ke terminal xinu
 - b. Jalankan perintah ps
 - c. Jalankan perintah namecmd

The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile". The terminal displays the XINU boot sequence, including the name "MUHAMMAD IKHSAN AL HAKIM" and the ASCII art logo "XINU". Below the logo, the text "Welcome to Xinu!" is shown. The user enters the command "ps", and the terminal displays a table of processes. The table has columns for Pid, Name, State, Prio, Ppid, Stack, Base, Stack Ptr, Stack Size, Msg, and Content. The processes listed are prnull, ipout, netin, Main process, shell, and ps. The user then enters the command "namecmd", and the terminal displays "My new command".

```
-----MUHAMMAD IKHSAN AL HAKIM-----
XINU
-----2311104064-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FF8 0x005F79AC 8192 0 0
6 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

- d. Jalankan perintah ps

```
Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE08 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 hello           wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- e. Lihat nama proses telah berubah

Nama proses berubah dari *netin* menjadi *hello* karena syscall `chname()` berhasil dijalankan melalui command `namecmd`.